



J D College Of Engineering & Management

Unit-4

Data Flow Testing

Dr. A. R. Welekar
Associate Professor(CSE)

❖ Data Flow Testing:

Introduction:

- Data Flow Testing is a type of structural testing . It is a method that is used to find the test paths of a program according to the locations of definitions and uses of variables in the program. It has nothing to do with data flow diagrams. Furthermore, it is concerned with:
 - ✓ Statements where variables receive values,
 - ✓ Statements where these values are used or referenced.
- By analyzing control flow graphs, this technique aims to identify issues such as unused variables or incorrect definitions, ensuring proper handling of data within the code.

❖ Data Flow Testing:

Introduction:

- To illustrate the approach of data flow testing, assume that each statement in the program is assigned a unique statement number. For a statement number S-

$$\text{DEF}(S) = \{X \mid \text{statement } S \text{ contains the definition of } X\}$$

$$\text{USE}(S) = \{X \mid \text{statement } S \text{ contains the use of } X\}$$

- If a statement is a loop or if condition then its DEF set is empty and the USE set is based on the condition of statement s.

❖ Data Flow Testing:

Introduction:

- Data Flow Testing uses the control flow graph to find the situations that can interrupt the flow of the program.
- Reference or defined anomalies in the flow of the data are detected at the time of associations between values and variables. These anomalies are:
 - ✓ A variable is defined but not used or referenced,
 - ✓ A variable is used but never defined,
 - ✓ A variable is defined twice before it is used

❖ Data Flow Graph:

Introduction:

- A Data Flow Graph (DFG) is a graphical representation that shows:
 - ✓ How data moves through a program
 - ✓ Where variables are defined (assigned values)
 - ✓ Where variables are used
 - ✓ The paths connecting definitions and uses

It is mainly used in Data Flow Testing

❖ Data Flow Graph:

Basic Components of Data Flow Graph:

Component	Meaning
Node	Represents a statement or block of code
Edge	Represents flow of control
Definition (def)	Where a variable gets value
Use	Where variable is used
d-u path	Path from definition to use

There are two types of uses:

- ✓ c-use (Computational use) → Used in calculation
- ✓ p-use (Predicate use) → Used in condition

❖ Data Flow Graph:

Example:

Step 1: Identify Definitions and Uses

Line	Variable	Type
2	a	def
3	b	def
4	a, b	p-use
5	c	def
5	a, b	c-use
7	c	def
7	a, b	c-use
8	c	c-use

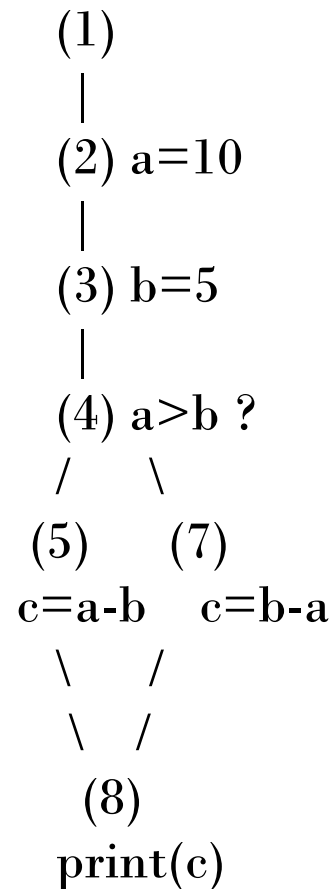
Simple Example Program

```
1. int a, b, c;  
2. a = 10;  
3. b = 5;  
4. if (a > b)  
5.   c = a - b;  
6. else  
7.   c = b - a;  
8. print(c);
```

❖ Data Flow Graph:

Simple Example Program

Step 2: Draw Data Flow Graph



```
1. int a, b, c;  
2. a = 10;  
3. b = 5;  
4. if (a > b)  
5.   c = a - b;  
6. else  
7.   c = b - a;  
8. print(c);
```


❖ Data Flow Graph:

Step 3: Data Flow Paths

Definition → Use paths for variable c

Path 1:

5 → 8

Path 2:

7 → 8

Simple Example Program

```
1. int a, b, c;  
2. a = 10;  
3. b = 5;  
4. if (a > b)  
5.   c = a - b;  
6. else  
7.   c = b - a;  
8. print(c);
```

❖ Data Flow Graph:

Why Data Flow Graph is Important:

It helps to find:

- ✓ Use of uninitialized variables
- ✓ Variables defined but not used
- ✓ Variables used but not defined
- ✓ Logical errors

Example of Error Detection

```
1. int x;  
2. int y = x + 5;
```

Here:

- ✓ x is used
- ✓ but never defined

❖ Data Flow Testing Criteria:

Data Flow Testing Criteria are the rules used to design test cases based on the definition and use of variables in a program.

These criteria ensure that every variable definition and its use is properly tested.

Main Data Flow Testing Criteria:

There are 5 important criteria:

- ✓ All Definitions (All-defs)
- ✓ All c-uses
- ✓ All p-uses
- ✓ All Uses
- ✓ All du-paths

❖ Data Flow Testing Criteria:

1. All-Definitions Criteria:

Definition: For every variable definition, at least one use must be tested

Variable	Defined at
x	Line 2
y	Line 3
z	Line 5,7

```
1. int x, y;  
2. x = 10;  
3. y = 5;  
4. if(x > y)  
5.   z = x + y;  
6. else  
7.   z = x - y;  
8. print(z);
```

❖ Data Flow Testing Criteria:

2. All c-uses Criteria:

Definition: Test all paths from definition to c-use

3. All p-uses Criteria:

Definition: Test all paths from definition to p-use

4. All Uses Criteria:

Definition: Test all c-uses and p-uses

Combination of:

- All c-uses
- All p-uses

5. All du-paths Criteria:

Definition: Test every path from definition to use

```
1. int x, y;  
2. x = 10;  
3. y = 5;  
4. if(x > y)  
5.   z = x + y;  
6. else  
7.   z = x - y;  
8. print(z);
```